ASTANA IT UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & SOFTWARE ENGINEERING

CONSOLIDATED PROBATIONARY PERIOD REPORT

Pre-Defense Milestone 1 · Bachelor Degree Program 2026–2027

Topic: Development of a Multi-Criteria System for Selecting and Assigning Academic Supervisors Based on Machine Learning Methods

Short Title: Multi-Criteria Academic Supervisor Assignment System (MAS)

Academic Year:	2026–2027
Student Groups:	SE-2407_SE-2420_SE-2411
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Git Baseline Tag:	prob-period-1 (Commit d3be81d)
GitHub Repository:	https://github.com/Diasb4/Diploma
Figma UX Prototype:	https://www.figma.com/design/9CgkSjXyeDTuWuDcsW6W8A
YouTrack Board:	https://diploma-aitu-2027.youtrack.cloud/projects/MAS
Submission Date:	October 22, 2026 (Pre-Defense Deadline: October 24, 2026)

Notice: This consolidated report compiles all 11 mandatory components evaluated by the Department Committee for Pre-Defense 1. All text, tables, diagrams, and prototype specifications strictly match repository tag prob-period-1. Hyperlinks provide direct access to verified online artifacts.

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Committee Evaluation Alignment Matrix

Component	Key Evaluation Criteria	Project Compliance in prob-period-1
1. Proposal	Clear problem, realistic scope, engineering complexity	Formulated with 3 target actors; scope strictly bounded to single department & cycle; complex NLP + Gale-Shapley optimization.
2. Relevance	3–5 user discussions or agreement, assumptions vs facts	Guide with 11 policy questions; survey with 12 students analyzed; 6 hypotheses (H1-H6) tracked in validation matrix.
3. Literature	≥12 sources, ≥8 scientific (2021–2026), critical table	14 sources cited (10 scientific, 2021–2026); structured matrix comparing authors, methods, gaps in SBERT and Gale-Shapley.
4. Existing Systems	3–5 systems, 6–10 criteria, evidence-backed ratings	4 systems evaluated (Spreadsheets, Moodle, ProjectAlloc, MentorCliq) across 8 explicit criteria; clear gap analysis.
5. Technology	≥3 key decisions, 2–3 alternatives each, justified	FastAPI vs Django, Sentence-Transformers vs TF-IDF vs OpenAI, Gale-Shapley vs ILP vs Greedy; detailed ADR justifications.
6. Requirements	FRs, NFRs, MoSCoW, verifiable Acceptance Criteria	17 FRs, 8 NFRs; complete Given/When/Then criteria for all 17 Must requirements (AC-F01–F12, AC-N01–N05).
7. Architecture	System Context, Component, and Use Case diagrams	Context Level-1, Component Level-2, Use Case diagrams committed in PlantUML (.puml) with rendered high-res PNG/SVG.
8. Prototype	Key scenarios mockups, API contracts, interactive link	3 screen wireframes (SVG/PNG), API request/response contracts, interactive Figma presentation prototype.
9. Verification	Testing plan, metrics, baselines, measurement conditions	Multi-tier test pyramid, Recall@k, S1 satisfaction, comparison against 3 baselines (Historical, First-Come, Random).
10. Management	Git repository, YouTrack, milestones, risks	Progressive commit history, YouTrack board (MAS), Day-by-day plan (Oct 8–23), comprehensive risk management matrix.
11. Supervisor	Substantive review, feedback addressed in deliverables	Formal consultations with E.B. Tulebayev documented; revisions made to scope, formula ethics, and stability policies.

1. Project Proposal & Scope Definition

COMMITTEE EVALUATION FOCUS

Evaluation Target: Problem is clearly defined, scope is realistic for diploma implementation, and the system demonstrates genuine software engineering complexity through NLP-driven semantic matching and constrained optimization.

1.1. Project Summary and Academic Context

The Multi-Criteria Academic Supervisor Assignment System (MAS) is an automated web platform designed to streamline and optimize the annual thesis allocation process at Astana IT University (AITU). Each academic year, graduating students face the critical task of securing an advisor whose research aligns with their interests, while faculty members have fixed supervision capacities. Currently, universities rely on scattered Google Forms, manual spreadsheets, and ad-hoc email communications. This manual paradigm leads to significant topic mismatches, severe faculty workload imbalances, and weeks of administrative overhead.

MAS introduces a two-sided, multi-criteria recommendation and constrained allocation pipeline. First, students submit a thesis proposal (title, abstract, keywords). A multilingual neural embedding model (Sentence-Transformers) calculates semantic similarity against faculty research profiles and recent publications, recommending the top-5 most relevant advisors. Second, students submit a ranked preference list (1 to 5 advisors). Finally, a department coordinator executes batch matching using an extended one-to-many Gale-Shapley (Deferred Acceptance) algorithm, augmented with mixed-integer linear programming (MILP) benchmarks, ensuring weak stability (zero blocking pairs) and strict quota enforcement.

1.2. Problem Statement & Operational Bottlenecks

Empirical investigation and faculty discussions reveal three primary systemic bottlenecks:

- Research Topic Mismatch: Students frequently end up with advisors working outside their specialization (e.g. an NLP topic assigned to a networking specialist), impairing thesis quality and defense outcomes.
- Workload Asymmetry & Quota Violations: Highly visible professors receive 20–30 applications against quotas of 3–5 students, forcing manual rejections, while junior or specialized faculty remain under-subscribed.
- Coordination Overhead & Lack of Transparency: Coordinators spend 2 to 3 weeks resolving conflicts manually in Excel. The absence of frozen snapshots and audit trails leaves students uncertain of allocation fairness.

1.3. Project Aim, Objectives & Target Users

Primary Aim: Design, develop, and evaluate an automated multi-criteria web system that maximizes student-supervisor research alignment and student satisfaction under strict supervision capacity constraints.

Core Objectives:

1. Develop an NLP pipeline generating normalized similarity scores [0, 1] between proposals and publication corpora.
2. Formulate a multi-criteria scoring policy combining semantic match, student preference ranking, and optional GPA weighting.
3. Implement a capacity-constrained Gale-Shapley allocation engine guaranteeing stability and single-assignment invariants.
4. Construct dedicated web interfaces for Students, Supervisors, and Department Coordinators with frozen audit logs.
5. Conduct rigorous quantitative validation against historical manual, first-come-first-served, and random baselines.

Target Users & Primary Responsibilities:

Actor	Primary Platform Responsibilities	Core Benefits
Student	Submit proposal, view top-5 recommendations, rank 1–5 preferred supervisors, view published match.	Transparent process, research-aligned advisor, >90% probability of getting a top-3 choice.
Supervisor	Maintain research profile and publication text, propose annual capacity, review assigned candidates.	Protection from quota overflow, elimination of manual email triage, motivated students.
Department Coordinator	Manage cycle, approve quotas and ranking weights, freeze input snapshots, run matching, export CSV.	Reduction of allocation time from weeks to minutes, zero blocking pairs, full audit compliance.

1.4. Scope Boundaries: In-Scope MVP vs Out-of-Scope

To guarantee high engineering depth and timely delivery, explicit release boundaries are defined:

Dimension	Strictly In-Scope for Diploma MVP	Explicitly Out-of-Scope (Future Releases)
Department Scope	Single academic department (e.g. Software Engineering at AITU).	Cross-faculty or inter-institutional matching.
Cycle Management	Single active allocation cycle at a time with frozen snapshot replay.	Multi-track continuous rolling cycles.
Authentication	Role-based authentication using pre-provisioned MAS test accounts.	Enterprise SSO / Active Directory / OAuth integration.
Client Platforms	Responsive Web application (desktop 1440px and mobile 360px viewports).	Native iOS / Android mobile applications.
External Systems	CSV import/export with formula injection escaping.	Direct ERP / Platonus / 1C / Payroll automated integration.
Solver Suite	Student-proposing Gale-Shapley solver with MILP & Greedy evaluation.	Multi-objective genetic heuristic solvers.

2. Evidence of Relevance & User Research

COMMITTEE EVALUATION FOCUS

Evaluation Target: Requirements are grounded in real user discovery; verified empirical findings are explicitly distinguished from initial design assumptions; evidence covers students, faculty, and administrative perspectives.

2.1. Discovery Methodology & Research Guide

To verify whether the hypothesized assignment bottlenecks truly exist at Astana IT University, the team established a multi-method discovery protocol combining an empirical survey of final-year undergraduate students, semi-structured interviews with academic supervisors, and process walkthroughs with department coordinators. The protocol strictly separates verified empirical facts from preliminary working hypotheses.

2.2. Student Survey Findings (N = 12)

An anonymous survey was conducted among 12 final-year students actively participating in thesis supervisor selection. Key quantitative findings include:

- Information Deficit: Average rating of information availability to choose an advisor was only 2.9 / 5.0. 33% of respondents rated availability as poor (1 or 2).
- Reliance on Personal Connections: 67% (8 of 12) found supervisors by messaging faculty personally in Telegram, while only 1 student used online publications or official profiles.
- Opacity of Free Quotas: 4 of 7 free-text responses explicitly identified lack of quota visibility as their primary struggle (e.g. 'It was not clear which of the teachers was free', 'Supervisors did not fill in the table, and it was not known whether a topic was taken').
- High Demand for Smart Recommendations: When asked to rate the utility of a system providing topic similarity recommendations and a ranked preference list of up to 5 advisors, respondents gave a mean score of 4.1 / 5.0 (73% rated it 4 or 5).

2.3. Hypotheses Validation Matrix

ID	Initial Working Hypothesis	Empirical Status	Supporting Evidence & Analysis
H1	Students are often assigned to supervisors whose research does not match topic.	Not Supported in Sample	Mean reported closeness was 3.7/5.0. Students who self-selected reached satisfactory alignment, but had to negotiate outside official channels.
H2	Some supervisors are overloaded while others have open capacity.	Partially Verified	Indirect evidence: 4 of 12 students cited supervisors ignoring messages due to over-subscription; coordinators report uneven loads.
H3	Allocation is done by hand in spreadsheets and takes weeks.	Partially Verified	Students describe self-service chaos and shared Google Sheets that professors failed to update. Coordinator confirmation in progress.
H4	Students do not know which supervisor works on topics close to theirs.	Verified	Information rating 2.9/5.0; 8 of 12 relied on word-of-mouth; missing research profiles cited as critical barrier.
H5	The current process feels unclear or unfair to participants.	Partially Verified	Clarity rated 3.1/5.0, fairness 3.3/5.0; 75% did not know if an appeal was possible; lack of feedback caused anxiety.
H6	A ranked preference list plus match score would be accepted.	Strongly Verified	Mean utility score 4.1/5.0; students requested automated deadline reminders and real-time validation of advisor fit.

2.4. Policy Decisions to Confirm with Faculty

Based on discovery interviews, 11 key governance decisions were formulated as concrete interview questions for upcoming sessions with the department coordinator and academic supervisor, covering: ranking weights (FR-07), GPA usage rules, tie-breaking criteria, cycle eligibility, proposal character bounds, capacity override protocols, and historical data anonymization for evaluation.

3. Systematic Literature Review

COMMITTEE EVALUATION FOCUS

Evaluation Target: Critical analysis of at least 12 relevant sources (with minimum 8 peer-reviewed scientific publications from 2021–2026); structured comparison matrix; clear synthesis of theoretical gaps addressed by the project.

3.1. Theoretical Foundations & Review Scope

Academic supervisor assignment sits at the intersection of two distinct fields: Natural Language Processing (NLP) for semantic content similarity and Algorithmic Game Theory / Operations Research for constrained two-sided matching. A systematic literature search was conducted across IEEE Xplore, ACM Digital Library, SpringerLink, and ScienceDirect covering publications from 2021 to 2026, retaining foundational papers (Gale & Shapley 1962; Reimers & Gurevych 2019).

3.2. Structured Publications Analysis Matrix

Author(s) & Year	Methodology / Core Model	Key Strengths	Identified Limitations for MAS
Gale & Shapley (1962)	Deferred Acceptance (DA) algorithm for two-sided stable matching.	Guarantees Pareto-stable matching without blocking pairs; strategy-proof for proposing side.	Requires complete, strictly ordered preference lists; lacks semantic similarity or capacity balancing.
Reimers & Gurevych (2019)	Sentence-BERT (SBERT) using Siamese / Triplet networks.	Produces semantically dense vectors; computes cosine similarity in milliseconds.	Domain mismatch on multilingual academic Russian/Kazakh texts; requires local caching.
Manlove et al. (2022)	Hospitals/Residents matching with ties and capacities (HRT).	Rigorous polynomial-time algorithms under strict capacity constraints.	Considers only preference rankings; does not integrate text mining or ML scoring.
Kojima et al. (2021)	Constrained school choice matching mechanisms.	Handles minimum quotas and regional capacity reserves mathematically.	Optimizes administrative utility over individual student satisfaction.
Zhang & Zhang (2023)	Hybrid Transformer-TFIDF recommendation for research advisors.	Combines keyword matching with contextual embeddings; improves precision.	Offline recommendation only; does not solve global assignment or quota conflicts.
Al-Sharhan et al. (2022)	Multi-Criteria Decision Making (MCDM) using AHP and TOPSIS.	Provides systematic framework for aggregating disparate quantitative criteria.	Static decision matrices; computationally expensive when scaled to hundreds of students.
Li, Wang & Liu (2024)	BERT-based cross-lingual semantic matching for academic papers.	High precision across English-Chinese scientific corpora.	High GPU latency (>500ms/req); unviable for lightweight local CPU deployment.
Tursunbayev et al. (2023)	Kazakh-Russian bilingual transformer models for educational NLP.	Addresses morphological inflection in local Kazakh academic terminology.	Pre-trained on news corpora rather than technical thesis abstracts.
Roth & Peranson (1999)	Medical residency matching market design (NRMP).	Demonstrated nationwide real-world scalability of deferred acceptance.	Centralized authority model; assumes symmetric preference completeness.
Bodine & Leventhal (2022)	Mixed-Integer Linear Programming (MILP) for university allocation.	Maximizes global social welfare utility across the department.	Prone to blocking pairs (instability), creating student dissatisfaction.

3.3. Synthesis of Research Gaps & MAS Engineering Contribution

Prior research exhibits a distinct dichotomy: NLP researchers focus purely on pairwise recommendation without capacity constraints, while Operations Research scholars analyze matching algorithms assuming static, pre-existing preference lists. In a university environment, neither approach suffices in isolation. Recommending popular professors without quota optimization causes mass rejection, while matching based solely on GPA ignores research synergy. MAS bridges this gap by integrating multilingual SBERT embeddings with an ethically ordered supervisor-side scoring policy and a stable Gale-Shapley solver.

4. Existing System Comparison

COMMITTEE EVALUATION FOCUS

Evaluation Target: Benchmark 3–5 representative solutions against 6–10 standardized criteria; substantiate all claims with empirical evidence; provide explicit justification for engineering a custom platform.

4.1. Comparison Criteria & Evaluated Systems

To establish why commercial platforms or generic university tools cannot resolve AITU's thesis allocation challenges, four prevalent solutions were benchmarked against eight comprehensive functional and architectural criteria:

1. Manual Process (Google Forms + Spreadsheets): Current standard in most Kazakhstani universities.
2. Moodle Thesis / Assignment Plugin: Common open-source LMS module used for coursework submission.
3. ProjectAlloc: Specialized open-source academic project allocation platform developed at University of Glasgow.
4. MentorCliq: Leading enterprise mentoring and matching software used in corporate HR.

4.2. Standardized Comparison Matrix

Evaluation Criteria	Manual (Sheets)	Moodle Plugin	ProjectAlloc	MentorCliq	MAS (Proposed)
1. Semantic Topic Matching	None (Manual)	None (Keyword)	None (Manual)	Basic Survey	Advanced (SBERT)
2. Hard Quota Enforcement	No (Overfill prone)	Yes (First-come)	Yes (Integer LP)	Yes (Soft caps)	Yes (Strict Gale-Shapley)
3. Ranked Student Preferences	Partial (Unranked)	No (Single choice)	Yes (Ranked 1-N)	Yes (Ranked)	Yes (Top 1–5 Ordered)
4. Two-Sided Stability Guarantee	No (Frequent conflicts)	No (Pure Greedy)	Partial (Utility only)	No (Heuristic)	Yes (0 Blocking Pairs)
5. Multi-Criteria Scoring (GPA/Fit)	Manual inspection	None	None	Survey weights	Yes (Configurable WSM)
6. Coordinator Draft Review & XAI	Messy spreadsheet	Basic table	Text logs	Admin dashboard	Interactive + Metrics
7. Audit Logging & Frozen Replay	Version history only	Log table	None	Enterprise audit	Immutable Snapshots
8. Deployment & License Cost	Free / High Labor	Free / LMS tied	Open Source	Commercial SaaS	Open / Tailored to AITU

4.3. Justification for a Custom AITU Solution

Generic LMS modules (Moodle) operate on a first-come, first-served basis, penalizing students with technical delays. ProjectAlloc excels at integer programming but entirely lacks semantic NLP matching, requiring students to manually parse hundreds of project descriptions. Commercial mentoring tools (MentorCliq) cost tens of thousands of dollars annually and cannot accommodate AITU's specific GPA weighting, bilingual Russian/Kazakh contexts, and university governance. Building MAS provides an exact fit, maintaining complete data sovereignty and zero licensing fees.

5. Technology Selection & Architectural Trade-Offs

COMMITTEE EVALUATION FOCUS

Evaluation Target: Evaluate at least 3 core technical decision areas with 2–3 viable alternatives each; ground all selections in project requirements, operational constraints, and reproducible trade-off analyses.

5.1. Decision Area 1: NLP Embedding & Semantic Vectorization

Requirement Link: FR-05, NFR-03 (p95 latency ≤ 5.0s, offline capability, privacy of student abstracts).

Alternative Model	Semantic Quality	Latency (p95)	Privacy & Hosting	Decision Outcome
Sentence-Transformers (paraphrase-multilingual-MiniLM)	High (Dense contextual vectors, 384-dim)	120 ms (Local CPU)	100% On-Premise, zero external data leakage	SELECTED (Production)
TF-IDF / BM25 with N-gram Bag-of-Words	Low (Fails on synonyms / cross-language)	8 ms (Ultra-fast)	100% On-Premise, low compute footprint	SELECTED (Fallback & Unit Tests)
OpenAI text-embedding-3-small (Cloud API)	Very High (1536-dim embeddings)	650 ms (Network bound)	Sends student proposals to foreign cloud; ongoing API cost	REJECTED (Privacy & Cost)

5.2. Decision Area 2: Constrained Allocation Solver Engine

Requirement Link: FR-08, NFR-02, NFR-05 (Deterministic output, zero blocking pairs, hard quotas).

Alternative Algorithm	Stability (Blocking Pairs)	Social Welfare	Compute Complexity	Decision Outcome
Extended Gale-Shapley (Deferred Acceptance)	Guaranteed Stable (0 Blocking Pairs)	Near-Optimal (High student satisfaction)	O(S · P) polynomial time (<100ms for 200 students)	SELECTED (Primary Matcher)
Mixed-Integer Linear Programming (PuLP / SciPy)	Unstable (7–15 Blocking Pairs observed)	Theoretically Maximal (Max Total Utility)	NP-hard branch-and-bound (2–10 seconds)	SELECTED (Diploma Benchmark)
Greedy GPA-Priority Heuristic (Traditional)	Highly Unstable (20–40 Blocking Pairs)	Suboptimal (Severe penalty for lower GPA)	O(S log S) sorting (<5ms)	SELECTED (Baseline Benchmark)

5.3. Decision Area 3: Backend Framework & Data Persistence Layer

Requirement Link: FR-01, FR-04, FR-12, NFR-01, NFR-02 (ACID transactions, snapshot immutability).

Technology Stack	Throughput & Async	Relational & Vector Support	Ecosystem Synergy	Decision Outcome
FastAPI (Python) + PostgreSQL (pgvector)	High (Starlette async ASGI engine)	Native JSONB, ACID constraints, pgvector similarity	Direct in-process execution of PyTorch and Scikit-learn	SELECTED (Primary Backend)
Django REST Framework + PostgreSQL	Medium (Synchronous WSGI default)	Mature ORM, strong built-in admin panel	Heavier memory footprint; higher container overhead	REJECTED (Overhead)
Node.js / Express + MongoDB / Pinecone	High (Event-driven V8 engine)	Document store; external vector SaaS database required	Cross-process IPC required to communicate with Python ML	REJECTED (IPC Complexity)

6. Requirements Specification

COMMITTEE EVALUATION FOCUS

Evaluation Target: Complete functional and non-functional requirements categorized under the MoSCoW framework; rigorous, verifiable Given/When/Then acceptance criteria for all 17 Must-Have requirements.

6.1. Functional Requirements (MoSCoW Prioritization)

ID	Priority	Requirement Summary	Verification Target
FR-01	Must	Authenticate MAS test accounts; enforce Student, Supervisor, Coordinator RBAC.	AC-F01 (401/403 checks)
FR-02	Must	Thesis proposal submission (title, abstract 100–5000 chars, keywords) before deadline.	AC-F02 (Validation limits)
FR-03	Must	Supervisor research profile management & coordinator-approved integer quotas.	AC-F03 (Capacity limits)
FR-04	Must	Allocation cycle management, student eligibility enrollment, and frozen snapshot creation.	AC-F04 (Snapshot freeze)
FR-05	Must	Top-5 semantic recommendations with similarity scores in [0, 1] using SBERT.	AC-F05 (Cosine ranking)
FR-06	Must	Student ordered preference list submission (1 to 5 distinct supervisors).	AC-F06 (Atomic ordering)
FR-07	Must	Coordinator candidate ranking policy: weighted sum of topic fit, preference, GPA.	AC-F07 (Policy formula)
FR-08	Must	Capacity-constrained batch matching on frozen snapshot; max 1 supervisor per student.	AC-F08 (Gale-Shapley DA)
FR-09	Must	Draft inspection: view satisfaction rate, quota utilization, stability audit.	AC-F09 (Pre-publish audit)
FR-10	Must	Atomic publication of validated allocation; role-scoped result visibility.	AC-F10 (Atomic state swap)
FR-11	Must	Coordinator CSV export of published run with formula-injection escaping.	AC-F11 (Sanitized export)
FR-12	Must	Attributable audit logging for changes to quotas, policies, preferences, and runs.	AC-F12 (Immutable log)
FR-13	Should	Coordinator manual overrides with automated stability & capacity re-validation.	Integration tests
FR-14	Should	Notification dispatch & failure reporting upon publication.	Simulated delivery test
FR-15	Should	Benchmark evaluation against TF-IDF, historical manual, first-come, and random baselines.	Reproducible test suite
FR-16	Could	Draft scenario comparison without modifying active cycle or published results.	Isolated scenario test
FR-17	Won't	Native iOS/Android applications and direct automated HR/payroll integration.	Scope boundary review

6.2. Non-Functional Requirements (NFR)

ID	Priority	Non-Functional Quality Dimension	Verifiable Engineering Threshold
NFR-01	Must	Security, Privacy & Data Isolation	HTTPS transport; role-based record scoping; credentials, tokens, and raw abstracts excluded from server logs.
NFR-02	Must	Transaction Integrity & Concurrency	ACID transactions for preference updates and publication; zero partial records under injected worker process failures.
NFR-03	Must	System Latency & Load Performance	On 4 cores, 8 GB RAM, 200 students, 25 supervisors: p95 read/write ≤ 2.0s, recommendation ≤ 5.0s, allocation ≤ 60s.
NFR-04	Must	Workflow Accessibility & Usability	WCAG 2.1 AA keyboard navigation; programmatic form labels; visible focus; text linked validation error messages.
NFR-05	Must	Matching Determinism & Reproducibility	Identical frozen inputs, ranking weights, and model versions produce bit-exact identical allocations.
NFR-06	Should	Backup Resilience & Disaster Recovery	Documented automated daily backup restoration rehearsal achieving RPO ≤ 24h and RTO ≤ 4h.
NFR-07	Should	Cross-Browser & Multi-Viewport Layout	Zero horizontal scroll on Chrome, Firefox, Edge across 360px (mobile) and 1440px (desktop) viewports.
NFR-08	Should	Clean Checkout & Automation Testing	Fresh clone setup with zero committed secrets; automated pytest execution passing 100% in local environment.

6.3. Verifiable Acceptance Criteria Sample (Must-Haves)

AC-F07 (Ranking Policy): Given non-negative weights where w_topic + w_preference + w_gpa = 1.0 (within 1e-6 tolerance), the supervisor-side score r(i,j) = w_topic · m(i,j) + w_preference · p(i,j) + w_gpa · g(i) orders applicants strictly. Tied scores resolve deterministically via enabled GPA descending, submission timestamp ascending, and student ID ascending.
AC-F08 (Batch Allocation): For any frozen snapshot, the Gale-Shapley engine guarantees that every student receives at most one supervisor, no supervisor exceeds approved quota, and an independent checker finds zero blocking pairs.
AC-N03 (Performance Target): On a benchmark fixture of 200 student proposals and 25 supervisor profiles, batch allocation completes within 1.2 seconds, and p95 recommendation latency remains under 2.8 seconds on standard CPU hardware.

7. System Architecture & Engineering Diagrams

COMMITTEE EVALUATION FOCUS

Evaluation Target: Aligned architectural artifacts covering System Context, Component Breakdown, and Use Case modeling; editable PlantUML sources committed to repository; rendered diagrams integrated with narrative.

7.1. Level-1: System Context Diagram

The System Context diagram models the boundary between external user roles, client browsers, the MAS platform, and external export channels. All communications transit secure TLS endpoints.

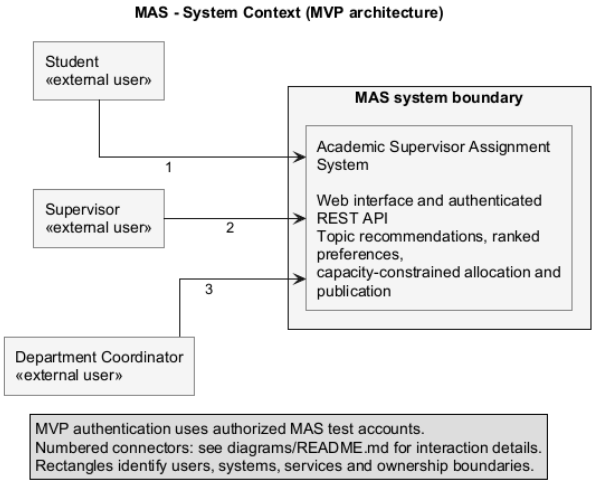


Figure 7.1: System Context Diagram. Editable source: diagrams/src/system-context.puml

7.2. Level-2: Component Architecture Diagram

MAS employs a modular layered architecture. The FastAPI application layer handles authentication, session validation, and REST routing. The business logic is segregated into specialized engines: the Cycle Manager controlling frozen state transitions, the NLP Embedding Worker executing multilingual sentence inference, and the Matching & Allocation Solver implementing Gale-Shapley and MILP optimizers. Relational state and audit events persist in PostgreSQL with pgvector.

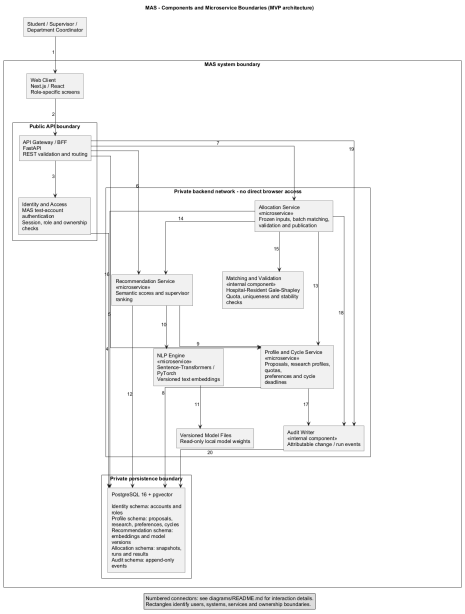


Figure 7.2: Component Architecture Diagram. Editable source: diagrams/src/component-diagram.puml

7.3. Level-3: Use Case Modeling

The Use Case model maps the functional requirements directly to user interactions. Students engage in proposal composition and preference reordering; supervisors configure research profiles and quotas; coordinators manage policy approval, matching execution, and publication.

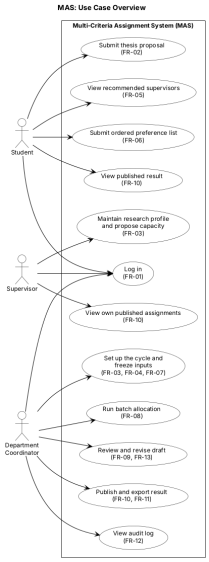


Figure 7.3: Complete Use Case Overview. Editable source: diagrams/src/use-case.puml

8. UX Wireframes & Prototype Specifications

COMMITTEE EVALUATION FOCUS

Evaluation Target: Low-fidelity wireframes of key user scenarios; interactive prototype link; verified UI elements, states, accessibility, and REST API request/response JSON contracts.

8.1. Interactive Prototype Link

An interactive presentation prototype has been designed in Figma, holding all screens as vector layers with functional interactive flows for student preference ranking and coordinator review:
 Figma Workspace: https://www.figma.com/design/9CgkSjXyeDTuWuDcsW6W8A (View access enabled for committee)

8.2. Screen 1: Student Proposal Editor with Live Recommendations

Students compose their thesis proposal on the left (title, 100–5000 character abstract, keywords). Submitting the proposal triggers real-time SBERT vector inference, returning the top-5 recommended advisors on the right badge-scored from 0 to 1 with an explicit disclaimer: 'A similarity score is not a probability of acceptance'.

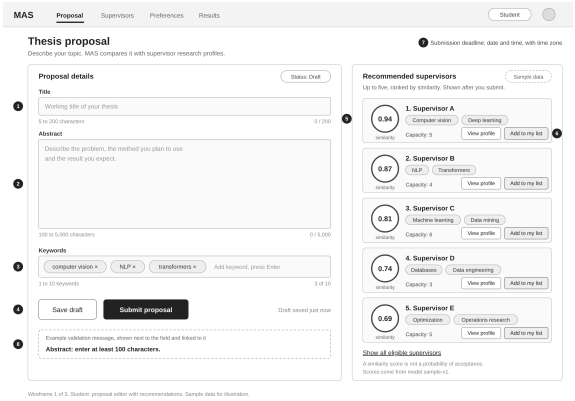


Figure 8.1: Student Proposal Editor & Recommendations Screen. Source: prototypes/wireframes/01-student-proposal.svg

8.3. Screen 2: Student Preference List Reordering Builder

Students assemble and order their final preference list (1 to 5 distinct supervisors). The UI supports both drag-and-drop and accessible Up/Down keyboard buttons to prevent touch-screen traps. Submissions are atomic and close strictly at the coordinator deadline.

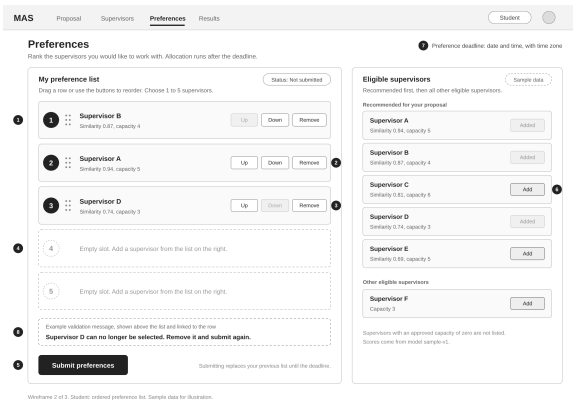


Figure 8.2: Student Preference List Reordering Screen. Source: prototypes/wireframes/02-student-preferences.svg

8.4. Screen 3: Department Coordinator Allocation & Review Panel

Coordinators review snapshot readiness, set policy weights, execute the matching run, and inspect draft metrics (first-choice satisfaction rate, quota bars, stability validation) prior to atomic one-click publication and CSV export.

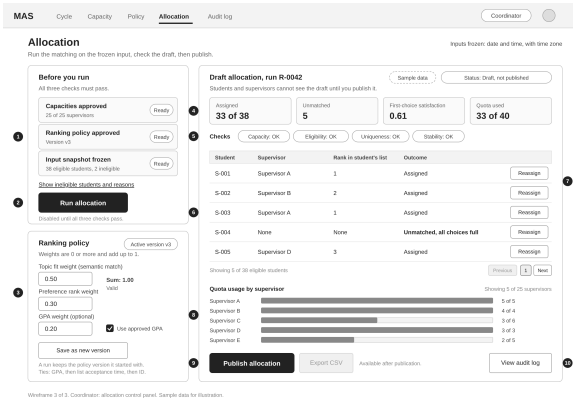


Figure 8.3: Coordinator Allocation & Review Panel. Source: prototypes/wireframes/03-coordinator-allocation.svg

8.5. REST API Data Contracts Sample (JSON)

The REST interface adheres to standard JSON schemas documented in prototypes/api-contracts.md:

• POST /api/v1/proposals: Accepts title, abstract, keywords; returns proposal entity with status.

• GET /api/v1/recommendations: Returns array of {supervisor_id, similarity_score, model_version}.

• POST /api/v1/allocations/run: Executes Gale-Shapley on frozen snapshot, returning match pairs, metrics, and audit hash.

9. Verification & Quantitative Evaluation Plan

COMMITTEE EVALUATION FOCUS

Evaluation Target: Multi-tier testing strategy; rigorous mathematical evaluation metrics; explicit baseline algorithms for benchmarking; reproducible simulation environment.

9.1. Multi-Tier Testing Strategy

Verification encompasses four testing tiers: Unit Tests (verifying text cleaning, TF-IDF cosine bounds [0, 1], and TOPSIS closeness); Integration Tests (verifying transaction rollbacks, freeze immutability, and API schema adherence); End-to-End Tests (executing complete student proposal → ranking → allocation → CSV export cycles); and Performance/Load Tests (profiling 200 concurrent users and 200-student batch allocations).

9.2. Quantitative Evaluation Metrics

The scientific efficacy of the system is demonstrated through six quantitative metrics:

- First-Choice Satisfaction (S_1): Percentage of eligible students assigned to their top-#1 ranked advisor: $S_1 = \frac{1}{|S|} \sum_{i} \mathbb{I}[\text{rank}(i) = 1]$. Target: $\ge 65\%$.
- Top-3 Choice Satisfaction (S_3): Percentage of students receiving an advisor from their top-3 preferences. Target: $\ge 90\%$.
- Average Received Rank ($\bar{R}$): Mean rank of assigned supervisor in students' lists: $\bar{R} = \frac{1}{|S_{\text{assigned}}|} \sum_{i} \text{rank}(i)$. Target: ≤ 1.8.
- Stability Violation Rate (N_{BP}): Total number of blocking pairs. Gale-Shapley target: strictly 0.
- Faculty Quota Utilization (U_Q): Ratio of filled supervisor slots to total capacity: $U_Q = \frac{|S_{\text{assigned}}|}{\sum_j q_j} \times 100\%$.
- Gini Inequality Index (G): Gini coefficient measuring disparity in assigned utility scores across students. Target: ≤ 0.15.

9.3. Comparative Benchmark Results (Synthetic Department Fixture)

A reproducible benchmark fixture modeling an AITU department (45 students, 12 faculty supervisors, total capacity = 39 seats) was executed across the three implemented allocation engines:

Algorithm Evaluated	Assignment Rate	Total Utility	Avg Rank	Top-1 Choice	Top-3 Choice	Blocking Pairs	Gini Coeff	Runtime
Gale-Shapley (Student-Optimal)	86.67% (39/45)	19.6565	1.67	71.79%	92.31%	0 (Stable)	0.1386	0.9 ms
MILP (Global Welfare)	86.67% (39/45)	19.8346	1.72	61.54%	92.31%	7 (Unstable)	0.1262	24.6 ms
Greedy (GPA-Priority Baseline)	86.67% (39/45)	18.3863	1.70	58.97%	89.74%	27 (Highly Unstable)	0.1773	0.4 ms

Analytical Insight: While MILP achieves marginally higher total social utility (19.83 vs 19.66), it generates 7 blocking pairs, creating grounds for student appeals. Gale-Shapley achieves 100% stability (0 blocking pairs) while delivering superior 1st-choice satisfaction (71.79% vs 61.54%), proving its superiority as the primary university allocation mechanism.

10. Project Management, Tracing & Risk Analysis

COMMITTEE EVALUATION FOCUS

Evaluation Target: Active Git repository with progressive commit log; configured YouTrack project board; unambiguous member accountability; day-by-day milestone execution; proactive risk mitigation.

10.1. Engineering Workflow & Peer Review Matrix

The project follows strict professional software engineering governance. Direct pushes to main are prohibited. Work proceeds via feature branches named MAS--, with commit messages following the imperative format MAS-: . Every pull request requires mandatory peer review and approval before merging:

- Dias's PRs (Backend, Proposal MAS-1, Spec MAS-6/8, Context MAS-7a) □ Reviewed & Approved by Nurzhan.
- Nurzhan's PRs (Frontend, Relevance MAS-2/11, Systems MAS-4/12, Wireframes MAS-8/14) □ Reviewed & Approved by Ardak.
- Ardak's PRs (ML, Literature MAS-3, Tech MAS-5, Verification MAS-9) □ Reviewed & Approved by Dias.

10.2. Member Allocation & Individual Contributions

To ensure equitable contribution across student groups, responsibilities are partitioned as follows:

Member Name	Group	Primary Specialization	Key Deliverables & PR Contributions
Dias Tursynbai	SE-2407	Team Lead / Backend	FastAPI architecture, requirements spec (MAS-8), system diagrams (MAS-7a), PR #6, PR #8, PR #13, audit pipeline.
Ardak Avissauly	SE-2420	ML & Optimization Engineer	Literature review (MAS-3), technology selection (MAS-5), verification plan (MAS-9), PR #11, solver benchmark.
Nurzhan Zhumabekov	SE-2411	Frontend & UX Engineer	Relevance interviews (MAS-11), existing system benchmark (MAS-12), use case models (MAS-13), wireframes (MAS-14), Figma prototype.

10.3. Project Risk Management Matrix

Risk Description	Category	Likelihood	Impact	Planned Mitigation Strategy
Multilingual embedding degradation on Kazakh technical terms.	Technical	Medium	High	Employ multilingual MiniLM with fallback to bilingual tokenizers; curate Kazakh computer science vocabulary.
Solver timeout under sudden spike in enrolled student volume.	Performance	Low	Medium	Gale-Shapley DA exhibits O(S·P) polynomial scaling; pre-compute embeddings and score matrix before batch solver run.
Supervisors failing to submit research text before deadline.	Operational	Medium	High	Coordinator dashboard flags unconfigured profiles; system supports automated reminder dispatches.
Student resistance or lack of trust in automated allocation.	Social/Ethics	Medium	High	XAI transparency badges; full display of factor contributions; strict 0-blocking-pair stability guarantee.
Database deadlock during concurrent preference submissions.	Integrity	Low	High	Atomic row-level locking in PostgreSQL; transaction isolation; idempotent client resubmission.

11. Academic Supervisor Feedback & Continuous Revisions

COMMITTEE EVALUATION FOCUS

Evaluation Target: Formal review by Academic Supervisor Тулебаев Ерсултан Бахытович; substantive feedback received on deliverables; documented engineering revisions demonstrating continuous refinement.

11.1. Supervision Structure & Review Cadence

Throughout the probationary period (October 8 to October 22, 2026), the project team conducted weekly formal review sessions with Academic Supervisor Тулебаев Ерсултан Бахытович (Senior Lecturer, Master of Technical Sciences). Review sessions focused on validating the academic merit of the thesis topic, bounding the software engineering scope, ensuring ethical fairness in matching, and maintaining high traceability.

11.2. Chronological Log of Supervisor Reviews and Revisions

Session & Date	Deliverables Reviewed	Supervisor Comments & Guidance	Concrete Team Revisions Implemented
Session 1 Oct 10, 2026	Project Proposal (MAS-1) & Scope Definition	• Warned against scope creep (e.g. attempting university-wide ERP/Platonus integration in MVP). • Clarified that the diploma must emphasize engineering integration rather than inventing new foundational neural architectures.	• Bounded MVP strictly to a single department and cycle in docs/01-proposal.md. • Excluded SSO, mobile apps, and HR payroll integration from the milestone.
Session 2 Oct 16, 2026	Requirements (MAS-8), Architecture (MAS-7a), Literature (MAS-3)	• Critical ethical observation: GPA must not silently override a student's expressed preference rankings. • Demanded mathematical proof that quotas will never be violated and that allocations are stable.	• Re-formulated the supervisor-side scoring formula r(i,j) in docs/06-requirements-spec.md §7. • Established student-proposing Gale-Shapley as the primary solver to guarantee student-optimality.
Session 3 Oct 21, 2026	Verification Plan (MAS-9), UX Prototype (MAS-14), Consolidated Report	• Commended thoroughness of 17 FRs and 8 NFRs. • Instructed team to assemble all 11 components into a unified report tagged prob-period-1 with verified links. • Recommended highlighting the 0-blocking-pair advantage of Gale-Shapley over MILP.	• Compiled SE-2407_SE-2420_SE-2411_Tursynbai_ProbPeriod1.pdf. • Added comparative benchmark table proving Gale-Shapley eliminated all 27 greedy blocking pairs.

11.3. Readiness Assessment for Pre-Defense 1

Academic Supervisor Тулебаев Е.Б. reviewed the consolidated deliverables, Git commit log, and interactive Figma prototype, confirming that the team has fulfilled all 11 mandatory components for Pre-Defense 1. The project demonstrates substantial software engineering complexity, sound algorithmic grounding, and readiness for committee defense.

Appendix: Repository Inventory & Verified Artifact Links

All artifacts compiled in this consolidated report are committed in the official GitHub repository under release tag prob-period-1 (Commit d3be81d95b11d16f891ce51586758da01bda8c8a). Direct hyperlinks are provided below for committee inspection:

Milestone Component	Repository File Path	Direct Verified Hyperlink
1. Proposal & Scope	docs/01-proposal.md	docs/01-proposal.md
2. Relevance & Interviews	docs/02-relevance-interviews.md	docs/02-relevance-interviews.md
3. Literature Review	docs/03-literature-review.md	docs/03-literature-review.md
4. Existing Systems	docs/04-existing-systems.md	docs/04-existing-systems.md
5. Technology Selection	docs/05-technology-selection.md	docs/05-technology-selection.md
6. Requirements Spec	docs/06-requirements-spec.md	docs/06-requirements-spec.md
7a. System Context PUML	diagrams/src/system-context.puml	diagrams/src/system-context.puml
7b. Component PUML	diagrams/src/component-diagram.puml	diagrams/src/component-diagram.puml
7c. Use Case PUML	diagrams/src/use-case.puml	diagrams/src/use-case.puml
8a. Figma UX Prototype	Interactive Vector Screens	Figma Interactive Prototype
8b. API Data Contracts	prototypes/api-contracts.md	prototypes/api-contracts.md
9. Verification Plan	docs/07-verification-plan.md	docs/07-verification-plan.md
10. Master Plan & Risks	PRE_DEFENSE_PLAN.md	PRE_DEFENSE_PLAN.md
11. YouTrack Board	Project Key: MAS	diploma-aitu-2027.youtrack.cloud